From the Bellman Equation to the Euler Equation

1 The Basic Idea

The Bellman equation and the Euler equation are two ways of describing the same dynamic optimization problem.

> **The Bellman equation describes the entire optimization problem. The Euler equation describes the optimal tradeoff between consumption today and consumption tomorrow.**

We can derive the Euler equation directly from the Bellman equation by combining a first-order condition with the envelope theorem.

2 The Bellman Equation

Consider an infinitely lived household that maximizes discounted lifetime felicity:

$$\sum_{t=0}^{\infty} \beta^t u(c_t), \qquad 0 < \beta < 1.$$

The household faces the budget constraint

$$c_t + a_{t+1} = Ra_t + y,$$

where a_t denotes beginning-of-period assets, $R = 1 + r$ is the gross interest rate, and y is constant labor income.

The Bellman equation is

$$V(a) = \max_{c,a'} \left\{ u(c) + \beta V(a') \right\}$$

subject to

$$c + a' = Ra + y.$$

Substituting the budget constraint into the objective,

$$\boxed{V(a) = \max_{a'} \left\{ u(Ra + y - a') + \beta V(a') \right\}.} \tag{1}$$

The value function $V(a)$ represents the maximum discounted lifetime felicity attainable with current assets a.

3 The First-Order Condition

Suppose the optimal choice of a' is interior and the value function is differentiable.

Differentiating the right-hand side of (1) with respect to a' gives

$$-u'(Ra + y - a') + \beta V'(a') = 0.$$

Since $c = Ra + y - a'$, this becomes

$$\boxed{u'(c) = \beta V'(a').} \tag{2}$$

The left-hand side is the marginal felicity sacrificed by saving one additional unit today.

The right-hand side is the discounted marginal value of entering the next period with one additional unit of assets.

At the optimum, these two quantities must be equal.

4 The Envelope Theorem

We now differentiate the Bellman equation with respect to current assets a.

By the envelope theorem, we can ignore the indirect effect of a on the optimal choice of a'.

Therefore,

$$\boxed{V'(a) = Ru'(c).} \tag{3}$$

The interpretation is straightforward: one additional unit of beginning-of-period assets provides R additional units of resources, each worth $u'(c)$ in marginal felicity.

Since the same relationship holds next period,

$$V'(a') = Ru'(c').$$

Substituting this expression into (2) yields

$$u'(c) = \beta R u'(c').$$

Restoring time subscripts, we obtain

$$\boxed{u'(c_t) = \beta(1 + r)u'(c_{t+1}).} \tag{4}$$

This is the **Euler equation**.

5 Economic Interpretation

Imagine that the household reduces consumption today by one unit and saves it instead.

The marginal cost is

$$\underbrace{u'(c_t)}_{\text{Felicity lost today}}.$$

The additional saving earns interest and provides $1 + r$ units of consumption tomorrow. Its discounted marginal benefit is

$$\underbrace{\beta(1+r)u'(c_{t+1})}_{\text{Discounted felicity gained tomorrow}}.$$

At an interior optimum, these must be equal:

$$\underbrace{u'(c_t)}_{\text{Marginal cost}} = \underbrace{\beta(1+r)u'(c_{t+1})}_{\text{Marginal benefit}}.$$

If the marginal benefit exceeded the marginal cost, the household would save more.
If the marginal cost exceeded the marginal benefit, the household would consume more today.
The Euler equation expresses the condition that neither adjustment can improve lifetime welfare.

6 An Example with CRRA Felicity

Suppose the felicity function is

$$u(c) = \begin{cases} \dfrac{c^{1-\gamma}-1}{1-\gamma}, & \gamma \neq 1, \\ \log c, & \gamma = 1. \end{cases}$$

Its marginal felicity is

$$u'(c) = c^{-\gamma}.$$

Substituting into the Euler equation,

$$c_t^{-\gamma} = \beta R c_{t+1}^{-\gamma}.$$

Rearranging,

$$\boxed{\frac{c_{t+1}}{c_t} = (\beta R)^{1/\gamma}.} \tag{5}$$

Thus,

- If $\beta R > 1$, consumption grows over time.
- If $\beta R = 1$, consumption is constant.
- If $\beta R < 1$, consumption declines over time.

The parameter γ determines how strongly consumption growth responds to the incentive to transfer resources across periods.

7 What Is the Difference?

The Bellman equation is a functional equation. Its solution gives both the value function and the optimal policy function.

The Euler equation is a necessary condition for an interior optimum. It describes how optimal consumption is related across adjacent periods.

The Euler equation alone is generally insufficient to determine the entire consumption path. We also need the budget constraints, initial conditions, and an appropriate terminal or transversality condition.

If a borrowing constraint binds, the Euler equality may be replaced by an inequality.

8 The Central Lesson

> **The Bellman equation tells us to choose the best possible future. The Euler equation tells us that, at the optimum, shifting a small amount of consumption between today and tomorrow cannot improve welfare.**

The connection between them is established by two steps:

$$\boxed{\text{Bellman equation} \xrightarrow{\text{First-order condition}} u'(c) = \beta V'(a')}$$

and

$$\boxed{\text{Envelope theorem} \implies u'(c_t) = \beta R u'(c_{t+1}).}$$